

BS (Data Science) Part-III
DASC- 523 Parallel and Distributed Computing

LAB # 4 – Parallel For loop in OpenMP

Lab Objective: To understand and work with the parallel execution of for loop in OpenMP.

Description:

- With a parallel for loop, threads split the range of iterations evenly and execute in parallel.
- For Example, there are 4 threads, and the For loop goes from 1 to 1000 meaning total of 1000 iterations. So, thread 1 will execute 1 to 250, thread 2 will execute 251 to 500, thread 3 will execute 501 to 750, and thread 4 will execute 751 to 1000.
- In this way, the time taken would reduce as the work is now not done in sequential order, rather it is executed in parallel.
- #pragma omp parallel for [clause[[,] clause] ...] new-line
for-loop

Task - Write a parallel for loop employing 4 threads for 10 loop iterations.

```
#include <omp.h>
#include <stdio.h>

int main()
{
    #pragma omp parallel for num_threads(4)
    for (int i = 1; i <= 10; i++) {
        int tid = omp_get_thread_num();
        printf("The thread %d executes i = %d\n", tid, i);
    }

    return 0;
}
```

Submitted
At 2/5/19
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